Samkedemia
EDUCATION

ULTIMATE SOLUTION BOOK:
ALL EXERCISES & PAST EXAMS

Mathematics for Business A

KU LEUVEN • BACHELOR OF BUSINESS ADMINISTRATION

VERSION: 4 NOV 2026

LET ME INTRODUCE MYSELF.

About Me

I'll make sure you pass this exam. I know exactly what it takes to get there.



BACKGROUND

- My name is **Jovan Samardzic**, I am 26 years old
- Got my **MSc in Statistics at KU Leuven**, graduated **Magna Cum Laude**
- Recipient of the **Science@Leuven Scholarship**

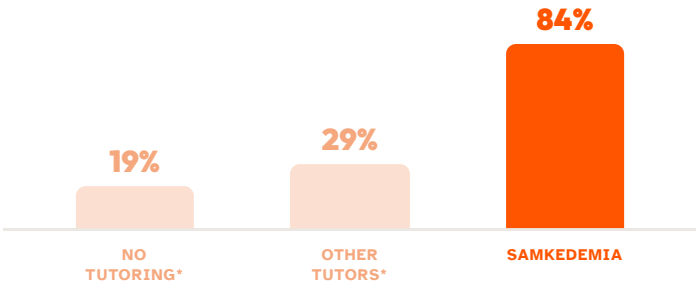
STYLE

- Founder of **Samkedemia**, helping students pass their exams
- Worked with **200+ university students** from all kinds of backgrounds
- **Plain language, not textbook theory** - just what actually makes sense

PROOF THAT IT WORKS.

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* BASED ON INTERNAL RESEARCH CONDUCTED IN 2026.

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Chapter 1 - Exponential and Logarithmic Functions

Question 1

Let x stand for any strictly positive number. Write the following expressions as a single power of x.

(a) $\left(x^{0.5}\right)^3$

(b) $x^2 \sqrt[3]{x}$

(c) $\left(\frac{1}{x^{0.25}}\right)^6$

(d) $\frac{x^3\sqrt{x}}{\sqrt[4]{x^3}}$

Tip: Exponent and Root Rules

The following rules let you rewrite any combination of powers and roots as a single power.

Power of a Power Rule: $(b^x)^y = b^{x \cdot y}$

Power of a Product Rule: $(a \cdot b)^x = a^x \cdot b^x$

Power of a Fraction Rule: $\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$

Product Rule: $b^x \cdot b^y = b^{x+y}$

Quotient Rule: $\frac{b^x}{b^y} = b^{x-y}$

Negative Exponent Rule: $b^{-x} = \frac{1}{b^x}$

Roots as Fractional Exponents: $\sqrt[n]{b} = b^{\frac{1}{n}}$

Solution 1: (a) $\left(x^{0.5}\right)^3$

Apply the power of a power rule:

$$\left(x^{0.5}\right)^3 = x^{0.5 \cdot 3} = x^{1.5} = x^{\frac{3}{2}}$$

Answer: The expression simplifies to $x^{\frac{3}{2}}$.

Solution 1: (b) $x^2 \sqrt[3]{x}$

Rewrite the cube root as a fractional exponent, then apply the product rule:

$$x^2 \sqrt[3]{x} = x^2 \cdot x^{\frac{1}{3}} = x^{2+\frac{1}{3}} = x^{\frac{7}{3}}$$

Answer: The expression simplifies to $x^{\frac{7}{3}}$.

Solution 1: (c) $\left(\frac{1}{x^{0.25}}\right)^6$

Rewrite the fraction using the negative exponent rule, then apply the power of a power rule:

$$\left(\frac{1}{x^{0.25}}\right)^6 = \left(x^{-0.25}\right)^6 = x^{-0.25 \cdot 6} = x^{-1.5} = x^{-\frac{3}{2}}$$

Answer: The expression simplifies to $x^{-\frac{3}{2}}$.

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Solution 1: (d) $\dfrac{x^3\sqrt{x}}{\sqrt[4]{x^3}}$

Rewrite both roots as fractional exponents, then apply the product and quotient rules:

$$\frac{x^3\sqrt{x}}{\sqrt[4]{x^3}} = \frac{x^3 \cdot x^{\frac{1}{2}}}{x^{\frac{3}{4}}} = x^{3+\frac{1}{2}-\frac{3}{4}} = x^{\frac{11}{4}}$$

Answer: The expression simplifies to $x^{\frac{11}{4}}$.

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Question 2

Simplify the following expressions.

(a) $\left(x^2y^3\right)^4$

(b) $x^{\frac{2}{3}}y^{\frac{1}{2}}x^{\frac{3}{4}}y^{\frac{1}{3}}$

(c) $\left(\frac{x}{yz}\right)^2\left(\frac{y}{xz}\right)^2\left(\frac{z}{xy}\right)^2$

Tip: Exponent and Root Rules

The following rules let you rewrite any combination of powers and roots as a single power.

Power of a Power Rule: $(b^x)^y = b^{x\cdot y}$

Power of a Product Rule: $(a\cdot b)^x = a^x\cdot b^x$

Power of a Fraction Rule: $\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$

Product Rule: $b^x\cdot b^y = b^{x+y}$

Quotient Rule: $\frac{b^x}{b^y} = b^{x-y}$

Negative Exponent Rule: $b^{-x} = \frac{1}{b^x}$

Roots as Fractional Exponents: $\sqrt[n]{b} = b^{\frac{1}{n}}$

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Solution 2: (a) $\left(x^2y^3\right)^4$

Apply the power of a product rule:

$$\left(x^2y^3\right)^4 = x^{2\cdot4}y^{3\cdot4} = x^8y^{12}$$

Answer: The expression simplifies to x^8y^{12}.

Solution 2: (b) $x^{\frac{2}{3}}y^{\frac{1}{2}}x^{\frac{3}{4}}y^{\frac{1}{3}}$

Group the terms with the same base, then apply the product rule to each base separately:

$$\begin{aligned} x^{\frac{2}{3}}y^{\frac{1}{2}}x^{\frac{3}{4}}y^{\frac{1}{3}} &= \left(x^{\frac{2}{3}}x^{\frac{3}{4}}\right)\left(y^{\frac{1}{2}}y^{\frac{1}{3}}\right) \\ &= x^{\frac{2}{3}+\frac{3}{4}}y^{\frac{1}{2}+\frac{1}{3}} \\ &= x^{\frac{8}{12}+\frac{9}{12}}y^{\frac{3}{6}+\frac{2}{6}} = x^{\frac{17}{12}}y^{\frac{5}{6}} \end{aligned}$$

Answer: The expression simplifies to $x^{\frac{17}{12}}y^{\frac{5}{6}}$.

Solution 2: (c) $\left(\frac{x}{yz}\right)^2\left(\frac{y}{xz}\right)^2\left(\frac{z}{xy}\right)^2$

Apply the power of a fraction rule to each factor, then combine the fractions and apply the

quotient rule:

$$\left(\frac{x}{yz}\right)^2\left(\frac{y}{xz}\right)^2\left(\frac{z}{xy}\right)^2 = \frac{x^2}{y^2z^2}\cdot\frac{y^2}{x^2z^2}\cdot\frac{z^2}{x^2y^2}$$

$$= \frac{x^2y^2z^2}{x^4y^4z^4}$$

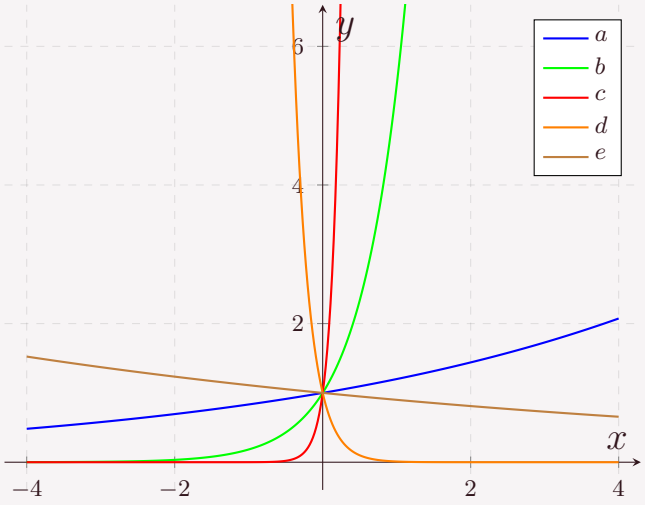
$$= x^{2-4}y^{2-4}z^{2-4} = x^{-2}y^{-2}z^{-2}$$

Answer: The expression simplifies to $x^{-2}y^{-2}z^{-2}$.

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Question 3

The graphs of five exponential functions are drawn in the following figure. The bases that were used are 2500, 0.01, 1.2, $2e$, and 0.9. Find out which graph corresponds to which exponential function.



Tip: Identifying Graphs of Exponential Functions

Exponential functions $y = a^x$ behave differently based on their base a:

- If $a > 1$, the graph increases as x increases. The larger the base, the faster it grows.
- If $0 < a < 1$, the graph decreases as x increases. The closer the base is to 0, the faster it decays.

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Solution 3: Match each graph to the base of its exponential function

Compare each curve's growth or decay speed to the five given bases, from the most extreme curves to the calmest ones.

Step 1: Find the fastest-growing curve.

Curve c shoots up almost vertically right at $x = 0$, reaching $y = 6$ before x even leaves 0. Since a larger base grows faster, this curve must belong to the largest base given, 2500.

Step 2: Find the fastest-decaying curve.

Curve d is the mirror image of curve c: it drops almost vertically to 0 just to the left of $x = 0$, then stays flat for $x > 0$. Since a base closer to 0 decays faster, this curve must belong to the base closest to 0, namely 0.01.

Step 3: Find the moderately fast-growing curve.

Curve b also grows as x increases, but far more gently than c - it only reaches $y = 6$ around $x \approx 1$. Of the two remaining bases above 1 (1.2 and $2e$), this faster growth matches the larger one, $2e \approx 5.4366$.

Step 4: Match the two remaining, slowly-changing curves.

Curve a increases only slowly across the whole picture, which matches the one remaining base above 1, namely 1.2. Curve e decreases just as slowly in the opposite direction, which matches the one remaining base below 1, namely 0.9.

Answer:

$$a \rightarrow \text{base } 1.2$$

$$b \rightarrow \text{base } 2e$$

$$c \rightarrow \text{base } 2500$$

$$d \rightarrow \text{base } 0.01$$

$$e \rightarrow \text{base } 0.9$$

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Question 4

A function f has an equation of the form:

$$y = A \cdot b^x,$$

where A and b are real numbers. The graph of f contains the points $(0.5, 1)$ and $(2.5, 9)$.

Determine the equation of f, i.e., find the values of A and b.

Tip: Solving for Two Unknowns by Dividing Two Equations

When two equations share the same two unknowns and each is a constant times a power with the same base, dividing one by the other cancels the common constant and leaves a single equation in one unknown:

$$\frac{c \cdot k^{t_1}}{c \cdot k^{t_2}} = k^{t_1 - t_2}$$

Solution 4: Find the equation of f

Step 1: Write one equation per given point.

Substitute each point into $y = A \cdot b^x$:

$$1 = A \cdot b^{0.5}$$

$$9 = A \cdot b^{2.5}$$

Step 2: Divide the two equations to eliminate A.

$$\frac{9}{1} = \frac{A \cdot b^{2.5}}{A \cdot b^{0.5}} = b^2$$

Step 3: Solve for b, then back-substitute for A.

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Take the square root of both sides:

$$b = (9)^{1/2} = 3$$

Substitute $b = 3$ into the first equation:

$$1 = A \cdot 3^{0.5} \quad \Rightarrow \quad A = \frac{1}{\sqrt{3}}$$

Step 4: Write the final equation of f.

$$f(x) = \frac{1}{\sqrt{3}} \cdot 3^x$$

Answer: $A = \frac{1}{\sqrt{3}}$ and $b = 3$, so $f(x) = \frac{1}{\sqrt{3}} \cdot 3^x$.

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Question 5

If possible, calculate the following without the use of a calculator.

(a) $\log_2 1$

(b) $\log_2 \left(\frac{1}{8}\right)$

(c) $\log_2 \left(\sqrt[4]{2^5}\right)$

(d) $\log_2 \left(\frac{1}{\sqrt[3]{2}}\right)$

(e) $\log_{\frac{1}{3}} 9$

(f) $\log_3 10$

(g) $\log(1\,000\,000)$

(h) $\log(10^{12})$

(i) $\log(0.001)$

(j) $\ln 1$

(k) $\ln \sqrt{e}$

(l) $\ln \left(\frac{1}{e}\right)$

Tip: Logarithm Rules

The following rules apply to logarithms in any base b.

Product Rule: $\log_b(x \cdot y) = \log_b(x) + \log_b(y)$

Quotient Rule: $\log_b \left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$

Power Rule: $\log_b(x^r) = r \cdot \log_b(x)$

Change of Base: $\log_b(x) = \frac{\ln(x)}{\ln(b)}$

Logarithm of the Base: $\log_b(b) = 1$

Logarithm of One: $\log_b(1) = 0$

Common Logarithm: $\log(x) = \log_{10}(x)$

Natural Logarithm: $\ln(x) = \log_e(x)$

Solution 5: (a) $\log_2 1$

Apply the logarithm of one rule:

$$\log_2 1 = 0$$

Answer: The expression evaluates to 0.

Solution 5: (b) $\log_2\left(\frac{1}{8}\right)$

Rewrite $\frac{1}{8}$ as a power of 2, then apply the power rule:

$$\log_2\left(\frac{1}{8}\right) = \log_2\left(2^{-3}\right) = -3 \cdot \log_2(2) = -3 \cdot 1 = -3$$

Answer: The expression evaluates to -3.

Solution 5: (c) $\log_2\left(\sqrt[4]{2^5}\right)$

Rewrite the fourth root as a fractional exponent, then apply the power rule:

$$\log_2\left(\sqrt[4]{2^5}\right) = \log_2\left(2^{\frac{5}{4}}\right) = \frac{5}{4} \cdot \log_2(2) = \frac{5}{4} \cdot 1 = \frac{5}{4}$$

Answer: The expression evaluates to $\frac{5}{4}$.

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Solution 5: (d) $\log_2\left(\frac{1}{\sqrt[3]{2}}\right)$

Rewrite the cube root and the fraction as a single power of 2, then apply the power rule:

$$\log_2\left(\frac{1}{\sqrt[3]{2}}\right) = \log_2\left(2^{-\frac{1}{3}}\right) = -\frac{1}{3}\cdot\log_2(2) = -\frac{1}{3}\cdot 1 = -\frac{1}{3}$$

Answer: The expression evaluates to $-\frac{1}{3}$.

Solution 5: (e) $\log_{\frac{1}{3}} 9$

Rewrite 9 as a power of $\frac{1}{3}$, then apply the power rule:

$$\log_{\frac{1}{3}}(9) = \log_{\frac{1}{3}}\left(\left(\frac{1}{3}\right)^{-2}\right) = -2\cdot\log_{\frac{1}{3}}\left(\frac{1}{3}\right) = -2\cdot 1 = -2$$

Answer: The expression evaluates to -2.

Solution 5: (f) $\log_3 10$

Since 10 cannot be written as a power of 3, this logarithm has no exact value that can be found without a calculator.

Answer: This cannot be computed exactly without a calculator.

Solution 5: (g) $\log(1\,000\,000)$

Rewrite $1\,000\,000$ as a power of 10, then apply the power rule:

$$\log(1\,000\,000) = \log(10^6) = 6 \cdot \log(10) = 6 \cdot 1 = 6$$

Answer: The expression evaluates to 6.

Solution 5: (h) $\log(10^{12})$

Apply the power rule directly:

$$\log(10^{12}) = 12 \cdot \log(10) = 12 \cdot 1 = 12$$

Answer: The expression evaluates to 12.

Solution 5: (i) $\log(0.001)$

Rewrite 0.001 as a power of 10, then apply the power rule:

$$\log(0.001) = \log(10^{-3}) = -3 \cdot \log(10) = -3 \cdot 1 = -3$$

Answer: The expression evaluates to -3.

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Solution 5: (j) $\ln 1$

Apply the logarithm of one rule:

$$\ln(1) = 0$$

Answer: The expression evaluates to 0.

Solution 5: (k) $\ln \sqrt{e}$

Rewrite the square root as a power of e, then apply the power rule:

$$\ln\left(\sqrt{e}\right) = \ln\left(e^{\frac{1}{2}}\right) = \frac{1}{2} \cdot \ln(e) = \frac{1}{2} \cdot 1 = \frac{1}{2}$$

Answer: The expression evaluates to $\frac{1}{2}$.

Solution 5: (l) $\ln\left(\frac{1}{e}\right)$

Rewrite the fraction as a power of e, then apply the power rule:

$$\ln\left(\frac{1}{e}\right) = \ln\left(e^{-1}\right) = -1 \cdot \ln(e) = -1 \cdot 1 = -1$$

Answer: The expression evaluates to -1.

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Question 6

Find the value of x:

(a) $\log_2 x = 4$

(b) $\log_x 49 = 2$

(c) $\ln(x+1) = 7$

Tip: Converting a Logarithmic Equation to Exponential Form

A logarithmic equation can always be rewritten in exponential form:

$$\log_b(x) = y \quad \Leftrightarrow \quad b^y = x$$

This works whichever of b, x, or y is the unknown: convert first, then solve.

Solution 6: (a) $\log_2(x) = 4$

Convert to exponential form:

$$x = 2^4 \quad \Rightarrow \quad x = 16$$

Answer: $x = 16$.

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Solution 6: (b) $\log_x(49) = 2$

Convert to exponential form and solve for x, keeping in mind that a logarithm's base must be positive:

$$49 = x^2 \quad \Rightarrow \quad x = \pm 7 \quad \Rightarrow \quad x = 7$$

Answer: $x = 7$ (the negative root is rejected since a logarithm's base must be positive).

Solution 6: (c) $\ln(x + 1) = 7$

Convert to exponential form:

$$x + 1 = e^7 \quad \Rightarrow \quad x = e^7 - 1 \approx 1095.6332$$

Answer: $x = e^7 - 1 \approx 1095.6332$.

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Question 7

Calculate (using the calculator):

(a) $\log(25)$

(b) $\log(40)$

(c) $\log(625)$

(d) Now explain:

i. why the sum of the logarithms in part (a) and (b) is equal to 3,

ii. why the logarithm in part (c) is twice the logarithm in part (a).

Solution 7: (a) $\log(25)$

Find the value using the calculator:

$$\log(25) \approx 1.3979$$

Answer: $\log(25) \approx 1.3979$.

Solution 7: (b) $\log(40)$

Find the value using the calculator:

$$\log(40) \approx 1.6021$$

Answer: $\log(40) \approx 1.6021$.

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Solution 7: (c) $\log(625)$

Find the value using the calculator:

$$\log(625) \approx 2.7959$$

Answer: $\log(625) \approx 2.7959$.

Tip: Logarithm Rules

The following rules apply to logarithms in any base b.

Product Rule: $\log_b(x \cdot y) = \log_b(x) + \log_b(y)$

Quotient Rule: $\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$

Power Rule: $\log_b(x^r) = r \cdot \log_b(x)$

Change of Base: $\log_b(x) = \frac{\ln(x)}{\ln(b)}$

Logarithm of the Base: $\log_b(b) = 1$

Logarithm of One: $\log_b(1) = 0$

Common Logarithm: $\log(x) = \log_{10}(x)$

Natural Logarithm: $\ln(x) = \log_e(x)$

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Solution 7: (d.i) Explain why $\log(25) + \log(40) = 3$

Apply the product rule to combine the two logarithms, then simplify the resulting power of 10:

$$\log(25) + \log(40) = \log(25 \cdot 40) = \log(1000) = \log(10^3) = 3 \cdot \log(10) = 3 \cdot 1 = 3$$

Answer: The sum equals 3 because $\log(25) + \log(40) = \log(1000) = \log(10^3) = 3$.

Solution 7: (d.ii) Explain why $\log(625) = 2 \cdot \log(25)$

Write 625 as a power of 25, then apply the power rule:

$$625 = 25^2 \quad \Rightarrow \quad \log(625) = \log(25^2) = 2 \cdot \log(25)$$

Answer: This holds because $625 = 25^2$, so $\log(625) = \log(25^2) = 2\log(25)$.

Question 8

Write as a single logarithm:

(a) $\ln 3 + \ln 7 - \ln 2 - 2 \cdot \ln 4$

(b) $0.5 \cdot (\log 225 + 8 \cdot \log 6 - 3 \cdot \log 169)$

Tip: Logarithm Rules

The following rules apply to logarithms in any base b.

Product Rule: $\log_b(x \cdot y) = \log_b(x) + \log_b(y)$

Quotient Rule: $\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$

Power Rule: $\log_b(x^r) = r \cdot \log_b(x)$

Change of Base: $\log_b(x) = \frac{\ln(x)}{\ln(b)}$

Logarithm of the Base: $\log_b(b) = 1$

Logarithm of One: $\log_b(1) = 0$

Common Logarithm: $\log(x) = \log_{10}(x)$

Natural Logarithm: $\ln(x) = \log_e(x)$

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Solution 8: (a) $\ln 3 + \ln 7 - \ln 2 - 2 \cdot \ln 4$

Apply the power rule to the last term, then combine everything into a single logarithm (added terms go in the numerator, subtracted terms in the denominator):

$$\ln 3 + \ln 7 - \ln 2 - 2\ln 4 \;=\; \ln 3 + \ln 7 - \ln 2 - \ln 16 \;=\; \ln\left(\frac{3 \cdot 7}{2 \cdot 16}\right) \;=\; \ln\left(\frac{21}{32}\right)$$

Answer: The expression simplifies to $\ln\left(\frac{21}{32}\right)$.

Solution 8: (b) $0.5 \cdot (\log 225 + 8 \cdot \log 6 - 3 \cdot \log 169)$

First, apply the power rule inside the parentheses to move the coefficients 8 and 3 into the exponent:

$$0.5 \cdot (\log 225 + 8\log 6 - 3\log 169) \;=\; 0.5 \cdot \Big(\log 225 + \log(6^8) - \log(169^3)\Big)$$

Next, combine the three terms into a single logarithm (added terms go in the numerator, subtracted terms in the denominator):

$$0.5 \cdot \Big(\log 225 + \log(6^8) - \log(169^3)\Big) \;=\; 0.5 \cdot \log\left(\frac{225 \cdot 6^8}{169^3}\right)$$

Now apply the power rule once more to bring the outer 0.5 inside as an exponent:

$$0.5 \cdot \log\left(\frac{225 \cdot 6^8}{169^3}\right) \;=\; \log\left(\left(\frac{225 \cdot 6^8}{169^3}\right)^{0.5}\right)$$

The exponent 0.5 means taking a square root, so simplify each factor separately:

$$\sqrt{225} \;=\; 15, \qquad \sqrt{6^8} \;=\; 6^4, \qquad \sqrt{169^3} \;=\; \left(13^2\right)^{\frac{3}{2}} \;=\; 13^3$$

Substituting these back gives the final simplified logarithm:

$$\log\left(\left(\frac{225\cdot 6^8}{169^3}\right)^{0.5}\right) = \log\left(\frac{15\cdot 6^4}{13^3}\right)$$

Answer: The expression simplifies to $\log\left(\frac{15\cdot 6^4}{13^3}\right)$.

Question 9

Calculate without using the graphical calculator:

(a) $\log 1 + \log 1000$

(b) $\log_7(7^8)$

(c) $\log_3\left(\frac{27}{81}\right)$

(d) $\ln e + \log\left(\frac{1}{10}\right)$

(e) $\log_6 54 - \log_6 9$

Tip: Logarithm Rules

The following rules apply to logarithms in any base b.

Product Rule:	$\log_b(x \cdot y) = \log_b(x) + \log_b(y)$
Quotient Rule:	$\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$
Power Rule:	$\log_b(x^r) = r \cdot \log_b(x)$
Change of Base:	$\log_b(x) = \frac{\ln(x)}{\ln(b)}$
Logarithm of the Base:	$\log_b(b) = 1$
Logarithm of One:	$\log_b(1) = 0$
Common Logarithm:	$\log(x) = \log_{10}(x)$
Natural Logarithm:	$\ln(x) = \log_e(x)$

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Solution 9: (a) $\log 1 + \log 1000$

Combine the two terms with the product rule, then rewrite as a power of 10 and apply the power rule:

$$\log 1 + \log 1000 = \log(1 \cdot 1000) = \log(10^3) = 3 \cdot \log(10) = 3 \cdot 1 = 3$$

Answer: The expression evaluates to 3.

Solution 9: (b) $\log_7(7^8)$

Apply the power rule, then evaluate $\log_7(7)$:

$$\log_7\left(7^8\right) = 8 \cdot \log_7(7) = 8 \cdot 1 = 8$$

Answer: The expression evaluates to 8.

Solution 9: (c) $\log_3\left(\frac{27}{81}\right)$

Simplify the fraction, rewrite it as a power of 3, then apply the power rule:

$$\log_3\left(\frac{27}{81}\right) = \log_3\left(3^{-1}\right) = -1 \cdot \log_3(3) = -1 \cdot 1 = -1$$

Answer: The expression evaluates to -1.

Solution 9: (d) $\ln e + \log\left(\frac{1}{10}\right)$

Rewrite $\frac{1}{10}$ as a power of 10, then apply the power rule to each term:

$$\ln(e) + \log\left(\frac{1}{10}\right) = 1 + \log\left(10^{-1}\right) = 1 + (-1)\cdot\log(10) = 1 + (-1) = 0$$

Answer: The expression evaluates to 0.

Solution 9: (e) $\log_6(54) - \log_6(9)$

Combine the two terms with the quotient rule, then apply the logarithm of the base rule:

$$\log_6(54) - \log_6(9) = \log_6\left(\frac{54}{9}\right) = \log_6(6) = 1$$

Answer: The expression evaluates to 1.

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Question 10

Compute $\log_3 27\underbrace{000\cdots000}_{\text{100 zeroes}}$, where the number in the logarithm consists of 27 followed by 100 zeroes.

Tip: Logarithm Rules

The following rules apply to logarithms in any base b.

Product Rule:	$\log_b(x \cdot y) = \log_b(x) + \log_b(y)$
Quotient Rule:	$\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$
Power Rule:	$\log_b(x^r) = r \cdot \log_b(x)$
Change of Base:	$\log_b(x) = \frac{\ln(x)}{\ln(b)}$
Logarithm of the Base:	$\log_b(b) = 1$
Logarithm of One:	$\log_b(1) = 0$
Common Logarithm:	$\log(x) = \log_{10}(x)$
Natural Logarithm:	$\ln(x) = \log_e(x)$

Solution 10: Compute $\log_3\left(27\underbrace{000\cdots000}_{\text{100 zeroes}}\right)$

Step 1: Split the logarithm with the product rule.

The number is 27 followed by 100 zeroes, i.e. $27 \cdot 10^{100}$:

$$\log_3\left(27 \cdot 10^{100}\right) = \log_3(27) + \log_3\left(10^{100}\right)$$

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Step 2: Evaluate the first term.

Since $27 = 3^3$, apply the power rule:

$$\log_3(27) \;=\; \log_3\left(3^3\right) \;=\; 3 \cdot \log_3(3) \;=\; 3$$

Step 3: Evaluate the second term.

Apply the power rule, then the change of base rule for $\log_3(10)$:

$$\log_3\left(10^{100}\right) \;=\; 100 \cdot \log_3(10) \;=\; 100 \cdot \frac{\ln(10)}{\ln(3)} \;\approx\; 100 \cdot 2.0959 \;=\; 209.5904$$

Step 4: Add the two results.

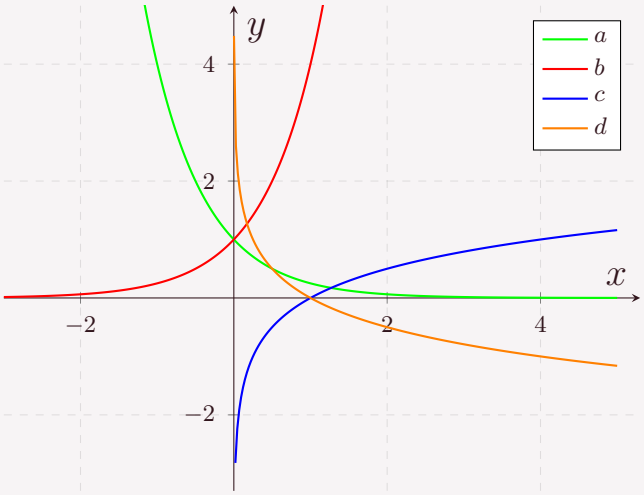
$$3 + 209.5904 \;=\; 212.5904$$

Answer: The value is approximately 212.5904.

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Question 11

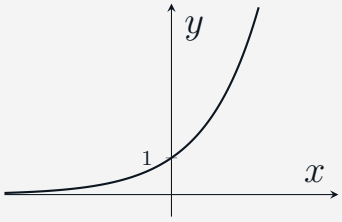
The graphs of two exponential and two logarithmic functions are drawn in the following figure. The bases that were used are 4 and 0.25. Write down the right function equation for each graph.



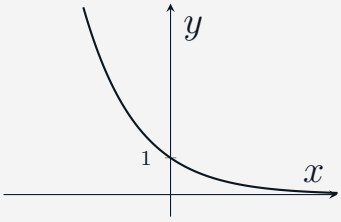
Tip: Graphs of Exponential Functions

The graph of $y = b^x$ is defined for every x. If $b > 1$ the graph rises as x grows; if $0 < b < 1$ it falls. Every such graph passes through $(0, 1)$.

$b > 1$ (rising)



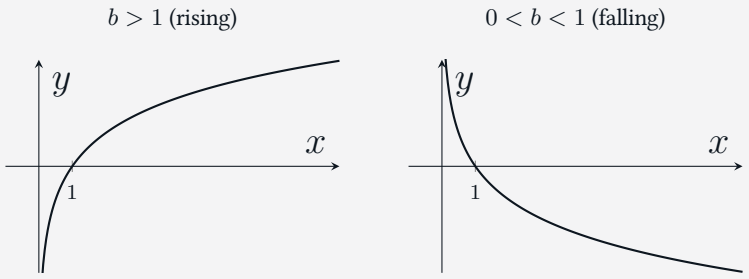
$0 < b < 1$ (falling)



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Tip: Graphs of Logarithmic Functions

The graph of $y = \log_b(x)$ is defined only for $x > 0$. If $b > 1$ the graph rises as x grows; if $0 < b < 1$ it falls. Every such graph passes through $(1, 0)$.



Solution 11: Match each graph to its function equation

Compare each curve's shape to the rules above, working from the broadest distinction down to the finest.

Step 1: Separate the exponential curves from the logarithmic curves.

Curves a and b have no vertical asymptote and are defined for every x: these are the two exponential functions. Curves c and d both have a vertical asymptote at $x = 0$ and are only defined for $x > 0$: these are the two logarithmic functions.

Step 2: Match the exponential curves to their bases.

Curve a decays as x increases, which matches the base below 1, namely 0.25. Curve b grows as x increases, which matches the base above 1, namely 4.

Step 3: Match the logarithmic curves to their bases.

Curve c increases slowly for $x > 0$, which matches the base above 1, namely 4. Curve d decreases for $x > 0$, which matches the base below 1, namely 0.25.

Step 4: Write down the equation for each graph.

Combining these matches gives the four function equations.

Answer:

$$a \rightarrow y = 0.25^x$$

$$b \rightarrow y = 4^x$$

$$c \rightarrow y = \log_4(x)$$

$$d \rightarrow y = \log_{0.25}(x)$$

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Question 12

In each of the following cases, rewrite the given expression.

(a) Rewrite $y = \log x$ in terms of $\ln x$.

(b) Rewrite $y = \log\left(\dfrac{1}{x}\right)$ in terms of $\log x$.

(c) Rewrite $y = \log\left(\dfrac{1}{x^2}\right)$ in terms of $\log x$.

(d) Rewrite $y = \ln x - \ln(x+3)$ as a single natural logarithm.

(e) Rewrite $y = \ln\sqrt{\dfrac{x^5(x-2)^8}{x-3}}$ in terms of $\ln x$, $\ln(x-2)$, and $\ln(x-3)$.

Tip: Logarithm Rules

The following rules apply to logarithms in any base b.

Product Rule:	$\log_b(x \cdot y) = \log_b(x) + \log_b(y)$
Quotient Rule:	$\log_b\left(\dfrac{x}{y}\right) = \log_b(x) - \log_b(y)$
Power Rule:	$\log_b(x^r) = r \cdot \log_b(x)$
Change of Base:	$\log_b(x) = \dfrac{\ln(x)}{\ln(b)}$
Logarithm of the Base:	$\log_b(b) = 1$
Logarithm of One:	$\log_b(1) = 0$
Common Logarithm:	$\log(x) = \log_{10}(x)$
Natural Logarithm:	$\ln(x) = \log_e(x)$

Solution 12: (a) $y = \log x$

Apply the change of base formula:

$$y = \log(x) = \frac{\ln(x)}{\ln(10)}$$

Answer: $y = \dfrac{\ln x}{\ln 10}$.

Solution 12: (b) $y = \log\left(\dfrac{1}{x}\right)$

Rewrite $\dfrac{1}{x}$ as x^{-1}, then apply the power rule:

$$y = \log\left(\frac{1}{x}\right) = \log\left(x^{-1}\right) = -\log(x)$$

Answer: $y = -\log x$.

Solution 12: (c) $y = \log\left(\dfrac{1}{x^2}\right)$

Rewrite $\dfrac{1}{x^2}$ as x^{-2}, then apply the power rule:

$$y = \log\left(\frac{1}{x^2}\right) = \log\left(x^{-2}\right) = -2\log(x)$$

Answer: $y = -2\log x$.

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Solution 12: (d) $y = \ln x - \ln(x+3)$

Apply the quotient rule:

$$y = \ln x - \ln(x+3) = \ln\left(\frac{x}{x+3}\right)$$

Answer: $y = \ln\left(\frac{x}{x+3}\right)$.

Solution 12: (e) $y = \ln\sqrt{\dfrac{x^5(x-2)^8}{x-3}}$

Rewrite the square root as an exponent of $\frac{1}{2}$, then apply the quotient, product, and power rules in turn:

$$\begin{aligned} y = \ln\sqrt{\frac{x^5(x-2)^8}{x-3}} &= \frac{1}{2}\ln\left(\frac{x^5(x-2)^8}{x-3}\right) \\ &= \frac{1}{2}\left(\ln(x^5) + \ln\left((x-2)^8\right) - \ln(x-3)\right) \\ &= \frac{1}{2}\left(5\ln x + 8\ln(x-2) - \ln(x-3)\right) \\ &= \frac{5}{2}\ln x + 4\ln(x-2) - \frac{1}{2}\ln(x-3) \end{aligned}$$

Answer: $y = \frac{5}{2}\ln x + 4\ln(x-2) - \frac{1}{2}\ln(x-3)$.

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Question 13

Solve for the variable:

(a) $\log_2(x) = 5 - \log_2(x+4)$

(b) $\log_2\left(\frac{2}{x}\right) = 3 + \log_2(x)$

(c) $3^{4x} = 9^{x+1}$

(d) $3 \cdot \left(\frac{1}{2}\right)^x + 7 = 55$

(e) $5 + 3 \cdot 4^{x-1} = 12$

(f) $2 \cdot 3^x - 5 \cdot \left(\frac{1}{3}\right)^x = 0$

(g) $\ln(2x-1) = 3$

(h) $5t^7 - 20 = 0$

(i) $\log(5x+1) = \log(4x+2)$

(j) $e^{3 \cdot \ln x} = 8$

(k) $\log(x^2 - x + 94) = 2$

(l) $\left|\log_{1/3}(x)\right| = 2$

(m) $2000 - 1560 \cdot 0.975^x > 800$

(n) $\frac{1}{7} \cdot 7^x \geq 49^{x+1}$

(o) $\left(\frac{2}{3}\right)^t < \frac{1}{30}$

(p) $0.96^{2x-1} > 4$

(q) $\sqrt{3^x} < \frac{1}{9}$

(r) $\left(\frac{1}{2}\right)^{1-x^2} \geq 8$

Tip: Solving Exponential and Logarithmic Equations and Inequalities

Exponential and logarithmic equations follow the same two-stage pattern: first check where the equation is defined, then solve it using one of the techniques below.

1. **Always check the domain first, before doing anything else.** For a logarithm $\log_b(A)$, the argument must satisfy:

$$A > 0$$

2. If a single logarithm equals a number, convert it to exponential form:

$$\log_b(x) = y \quad \Leftrightarrow \quad x = b^y$$

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3. If two logs with the same base are added or subtracted, combine them into one logarithm first:

$$\log_b(x) + \log_b(y) = \log_b(xy)$$

4. If two logs with the same base are set equal to each other, their arguments must be equal:

$$\log_b(x) = \log_b(y) \quad \Leftrightarrow \quad x = y$$

5. If you can write both sides of an exponential equation with the same base, compare the exponents directly:

$$b^x = b^y \quad \Leftrightarrow \quad x = y$$

6. If the bases cannot be matched, isolate the exponential term and take ln of both sides instead.

Two further techniques: the substitution $y = b^x$ turns an exponential equation into a polynomial one, and an absolute value equation splits into two cases:

$$|A| = k \quad \Leftrightarrow \quad A = k \text{ or } A = -k$$

Solution 13: (a) $\log_2(x) = 5 - \log_2(x+4)$

Step 1: Check the domain.

Both logarithms need a positive argument:

$$x > 0 \quad \text{and} \quad x + 4 > 0$$

The stricter condition is $x > 0$.

Step 2: Solve the equation.

Move both logarithm terms to one side and combine them with the product rule:

$$\log_2(x) + \log_2(x+4) = 5 \quad \Rightarrow \quad \log_2\big(x(x+4)\big) = 5$$

Convert to exponential form and solve the resulting quadratic:

$$x(x+4) \;=\; 2^5 \quad \Rightarrow \quad x^2+4x-32 \;=\; 0$$

$$\Rightarrow \quad (x+8)(x-4) \;=\; 0 \quad \Rightarrow \quad x=-8 \text{ or } x=4$$

Only $x=4$ satisfies the domain $x>0$.

Answer: $x=4$.

Solution 13: (b) $\log_2\left(\frac{2}{x}\right)=3+\log_2(x)$

Step 1: Check the domain.

Both logarithms need a positive argument:

$$\frac{2}{x} \;>\; 0 \quad \text{and} \quad x \;>\; 0$$

Both conditions come down to $x>0$.

Step 2: Solve the equation.

Expand the left side with the quotient rule and collect the logarithm terms:

$$\log_2(2)-\log_2(x) \;=\; 3+\log_2(x) \quad \Rightarrow \quad 1-\log_2(x) \;=\; 3+\log_2(x)$$

$$\Rightarrow \quad -2 \;=\; 2\log_2(x) \quad \Rightarrow \quad \log_2(x) \;=\; -1$$

Convert to exponential form:

$$x \;=\; 2^{-1} \;=\; 0.5$$

Answer: $x=0.5$.

Solution 13: (c) $3^{4x} = 9^{x+1}$

Step 1: Check the domain.

This is an exponential equation, defined for every real x - there is no domain restriction.

Step 2: Solve the equation.

Rewrite 9 as a power of 3 so both sides share the same base:

$$3^{4x} = (3^2)^{x+1} = 3^{2x+2}$$

Since the bases match, equate the exponents:

$$4x = 2x + 2 \quad \Rightarrow \quad x = 1$$

Answer: $x = 1$.

Solution 13: (d) $3 \cdot \left(\frac{1}{2}\right)^x + 7 = 55$

Step 1: Check the domain.

This is an exponential equation, defined for every real x - there is no domain restriction.

Step 2: Solve the equation.

Isolate the exponential term, then rewrite both sides as a power of 2:

$$3\left(\frac{1}{2}\right)^x = 48 \quad \Rightarrow \quad \left(\frac{1}{2}\right)^x = 16 \quad \Rightarrow \quad 2^{-x} = 2^4$$

$$\Rightarrow \quad -x = 4 \quad \Rightarrow \quad x = -4$$

Answer: $x = -4$.

Solution 13: (e) $5 + 3 \cdot 4^{x-1} = 12$

Step 1: Check the domain.

This is an exponential equation, defined for every real x - there is no domain restriction.

Step 2: Solve the equation.

Isolate the exponential term. Since 4 and $\frac{7}{3}$ don't share a common base, take $\ln$ of both sides instead:

$$3 \cdot 4^{x-1} = 7 \quad \Rightarrow \quad 4^{x-1} = \frac{7}{3}$$

$$\Rightarrow \quad (x-1)\ln 4 = \ln\left(\frac{7}{3}\right) \quad \Rightarrow \quad x = 1 + \frac{\ln(7/3)}{\ln 4} \approx 1.6112$$

Answer: $x = 1 + \dfrac{\ln(7/3)}{\ln 4} \approx 1.6112.$

Solution 13: (f) $2 \cdot 3^{x} - 5 \cdot \left(\frac{1}{3}\right)^{x} = 0$

Step 1: Check the domain.

This is an exponential equation, defined for every real x - there is no domain restriction.

Step 2: Solve the equation.

Substitute $y = 3^x$. The other power is then its reciprocal:

$$\left(\frac{1}{3}\right)^{x} = 3^{-x} = \frac{1}{y}$$

$$2y - \frac{5}{y} = 0 \quad \Rightarrow \quad 2y^2 - 5 = 0$$

$$\Rightarrow \quad y^2 = \frac{5}{2} \quad \Rightarrow \quad y = \sqrt{\frac{5}{2}}$$

(the negative root is rejected since $y = 3^x > 0$). Substitute back and take ln of both sides:

$$3^x = \sqrt{\frac{5}{2}} \quad \Rightarrow \quad x \ln 3 = \ln \sqrt{\frac{5}{2}}$$

$$\Rightarrow \quad x = \frac{\ln \sqrt{5/2}}{\ln 3} \approx 0.4170$$

Answer: $x = \dfrac{\ln \sqrt{5/2}}{\ln 3} \approx 0.4170.$

Solution 13: (g) $\ln(2x - 1) = 3$

Step 1: Check the domain.

We need $2x - 1 > 0$, i.e. $x > \dfrac{1}{2}$.

Step 2: Solve the equation.

Convert to exponential form:

$$2x - 1 = e^3 \quad \Rightarrow \quad x = \frac{e^3 + 1}{2} \approx 10.5428$$

This is comfortably greater than $\dfrac{1}{2}$, so the solution is valid.

Answer: $x = \dfrac{e^3 + 1}{2} \approx 10.5428.$

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Solution 13: (h) $5t^7 - 20 = 0$

Step 1: Check the domain.

This is a power equation, not an exponential one - an odd root is defined for every real t, so there is no domain restriction.

Step 2: Solve the equation.

Isolate t^7 and take a 7th root:

$$5t^7 = 20 \quad \Rightarrow \quad t^7 = 4 \quad \Rightarrow \quad t = \sqrt[7]{4} \approx 1.2190$$

Answer: $t = \sqrt[7]{4} \approx 1.2190.$

Solution 13: (i) $\log(5x + 1) = \log(4x + 2)$

Step 1: Check the domain.

Both logarithms need a positive argument:

$$5x + 1 > 0 \quad \Rightarrow \quad x > -0.2 \qquad\qquad 4x + 2 > 0 \quad \Rightarrow \quad x > -0.5$$

The stricter condition is $x > -0.2$.

Step 2: Solve the equation.

Apply the one-to-one property to equate the arguments:

$$5x + 1 = 4x + 2 \quad \Rightarrow \quad x = 1$$

This satisfies $x > -0.2$.

Answer: $x = 1.$

Solution 13: (j) $e^{3\cdot\ln(x)} = 8$

Step 1: Check the domain.

The term $\ln(x)$ requires $x > 0$.

Step 2: Solve the equation.

Apply the power rule to the exponent, then cancel e and $\ln$:

$$e^{\ln(x^3)} = 8 \quad \Rightarrow \quad x^3 = 8 \quad \Rightarrow \quad x = 2$$

Answer: $x = 2$.

Solution 13: (k) $\log(x^2 - x + 94) = 2$

Step 1: Check the domain.

The argument has a negative discriminant, so it has no zeros:

$$(-1)^2 - 4 \cdot 1 \cdot 94 = -375 < 0$$

So the argument is always positive. The domain is $x \in \mathbb{R}$.

Step 2: Solve the equation.

Convert to exponential form and solve the resulting quadratic:

$$x^2 - x + 94 = 10^2 \quad \Rightarrow \quad x^2 - x - 6 = 0$$

$$\Rightarrow \quad (x-3)(x+2) = 0 \quad \Rightarrow \quad x = 3 \text{ or } x = -2$$

Answer: $x = 3$ or $x = -2$.

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Solution 13: (l) $\left|\log_{1/3}(x)\right| = 2$

Step 1: Check the domain.

We need $x > 0$.

Step 2: Solve the equation.

Split into two cases, one for each sign, then convert each to exponential form:

$$\log_{\frac{1}{3}}(x) = 2 \quad \Rightarrow \quad x = \left(\frac{1}{3}\right)^{2} = \frac{1}{9}$$

$$\log_{\frac{1}{3}}(x) = -2 \quad \Rightarrow \quad x = \left(\frac{1}{3}\right)^{-2} = 9$$

Both values are positive, so both satisfy the domain.

Answer: $x = \dfrac{1}{9}$ or $x = 9$.

Tip: Taking a Logarithm of an Inequality

Taking ln of both sides of an inequality is always safe, since ln is increasing and keeps the inequality direction. Dividing afterwards can change it: dividing by $\ln(b)$ with $b > 1$ keeps the direction, but dividing by $\ln(b)$ with $0 < b < 1$ is dividing by a negative number, so the inequality sign flips.

Solution 13: (m) $2000 - 1560 \cdot 0.975^x > 800$

Step 1: Check the domain.

This is an exponential expression, defined for every real x - there is no domain restriction.

Step 2: Solve the inequality.

Isolate the exponential term:

$$-1560 \cdot 0.975^x > -1200 \quad \Rightarrow \quad 0.975^x < \frac{10}{13}$$

Take ln of both sides (safe, since ln is increasing), then divide by $\ln(0.975)$, which is negative, flipping the inequality:

$$x \ln(0.975) < \ln\left(\frac{10}{13}\right) \quad \Rightarrow \quad x > \frac{\ln(10/13)}{\ln(0.975)} \approx 10.3628$$

Answer: $x > \dfrac{\ln(10/13)}{\ln(0.975)} \approx 10.3628.$

Solution 13: (n) $\frac{1}{7} \cdot 7^x \geq 49^{x+1}$

Step 1: Check the domain.

This is an exponential inequality, defined for every real x - there is no domain restriction.

Step 2: Solve the inequality.

Rewrite both sides with base 7:

$$\frac{1}{7} \cdot 7^x = 7^{x-1} \qquad \text{and} \qquad 49^{x+1} = (7^2)^{x+1} = 7^{2x+2}$$

Since the base $7 > 1$, the inequality between the exponents keeps the same direction:

$$7^{x-1} \geq 7^{2x+2} \quad \Rightarrow \quad x - 1 \geq 2x + 2 \quad \Rightarrow \quad x \leq -3$$

Answer: $x \leq -3$.

Solution 13: (o) $\left(\frac{2}{3}\right)^t < \frac{1}{30}$

Step 1: Check the domain.

This is an exponential inequality, defined for every real t - there is no domain restriction.

Step 2: Solve the inequality.

Take ln of both sides, then divide by $\ln\left(\frac{2}{3}\right)$, which is negative, flipping the inequality:

$$t \ln\left(\frac{2}{3}\right) < \ln\left(\frac{1}{30}\right) \quad \Rightarrow \quad t > \frac{\ln(1/30)}{\ln(2/3)} \approx 8.3884$$

Answer: $t > \frac{\ln(1/30)}{\ln(2/3)} \approx 8.3884$.

Solution 13: (p) $0.96^{2x-1} > 4$

Step 1: Check the domain.

This is an exponential inequality, defined for every real x - there is no domain restriction.

Step 2: Solve the inequality.

Take ln of both sides, then divide by $\ln(0.96)$, which is negative, flipping the inequality:

$$(2x-1)\ln(0.96) > \ln(4) \quad \Rightarrow \quad 2x - 1 < \frac{\ln 4}{\ln 0.96}$$

$$\Rightarrow \quad x < \frac{1 + \ln 4/\ln 0.96}{2} \approx -16.4797$$

Answer: $x < \dfrac{1 + \ln 4/\ln 0.96}{2} \approx -16.4797.$

Solution 13: (q) $\sqrt{3^x} < \dfrac{1}{9}$

Step 1: Check the domain.

Since $3^x > 0$ for every real x, the square root is always defined - there is no domain restriction.

Step 2: Solve the inequality.

Rewrite both sides with base 3:

$$\sqrt{3^x} = 3^{x/2} \qquad \text{and} \qquad \frac{1}{9} = 3^{-2}$$

Since the base $3 > 1$, the inequality between the exponents keeps the same direction:

$$3^{x/2} < 3^{-2} \quad \Rightarrow \quad \frac{x}{2} < -2 \quad \Rightarrow \quad x < -4$$

Answer: $x < -4.$

Solution 13: (r) $\left(\dfrac{1}{2}\right)^{1-x^2} \geq 8$

Step 1: Check the domain.

This is an exponential inequality, defined for every real x - there is no domain restriction.

Step 2: Solve the inequality.

Rewrite both sides with base 2:

$$\left(\frac{1}{2}\right)^{1-x^2} = 2^{-(1-x^2)} \qquad \text{and} \qquad 8 = 2^3$$

Since the base $2 > 1$, the inequality between the exponents keeps the same direction:

$$2^{-(1-x^2)} \geq 2^3 \quad \Rightarrow \quad x^2 - 1 \geq 3$$

$$\Rightarrow \quad x^2 \geq 4 \quad \Rightarrow \quad x \leq -2 \text{ or } x \geq 2$$

Answer: $x \leq -2$ or $x \geq 2$.

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Question 14

John invests €10 000 in a savings account. The yearly (compound) interest rate is 1%. Find a formula for the compound amount S, i.e., the sum of the principal and all the interest earned after t years.

Tip: Compound Interest Formulas

Let $S(0)$ be the principal, r the annual interest rate as a decimal, and t the time in years.

$$\text{future value:} \quad S(t) = S(0) \cdot (1+r)^t$$

$$\text{total interest earned:} \quad S(t) - S(0)$$

$$\text{interest in year } t\text{:} \quad S(t) - S(t-1)$$

Solution 14: Find the formula for the compound amount $S(t)$

Before we start, we define the variables used in the exercise:

Variables:

$S(t)$	compound amount after t years
t	time in years

The principal is $S(0) = 10\,000$ and the annual interest rate is $r = 0.01$. Substitute these values into the compound interest formula:

$$S(t) = 10\,000 \cdot (1+0.01)^t = 10\,000 \cdot 1.01^t$$

Answer: $S(t) = 10\,000 \cdot 1.01^t$ euro.

Question 15

Suppose \$2 000 is invested at 3% compounded annually.

(a) Find the value of the investment after 5 years.

(b) Find the interest earned over the first 5 years.

(c) Find the interest earned during the 5th year only.

Tip: Compound Interest Formulas

Let $S(0)$ be the principal, r the annual interest rate as a decimal, and t the time in years.

$$\text{future value:} \quad S(t) = S(0) \cdot (1+r)^t$$

$$\text{total interest earned:} \quad S(t) - S(0)$$

$$\text{interest in year } t\text{:} \quad S(t) - S(t-1)$$

Solution 15: (a) Find the value of the investment after 5 years

Before we start, we define the variables used in the exercise:

Variables:

$S(t)$	value of the investment after t years
t	time in years

Substitute $S(0) = 2000$, $r = 0.03$, and $t = 5$ into the future value formula:

$$S(5) = 2000 \cdot (1.03)^5 \approx 2318.55$$

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Answer: The value after 5 years is approximately \$2318.55.

Solution 15: (b) Find the interest earned over the first 5 years

Using $S(5)$ from part (a), apply the total interest formula:

$$\text{Interest} \;=\; S(5) - S(0) \;=\; 2318.55 - 2000 \;=\; 318.55$$

Answer: The total interest earned is \$318.55.

Solution 15: (c) Find the interest earned during the 5th year only

Step 1: Find the value after 4 years.

$$S(4) \;=\; 2000 \cdot (1.03)^4 \;\approx\; 2251.02$$

Step 2: Apply the interest-in-a-specific-year formula.

$$\text{Interest in year } 5 \;=\; S(5) - S(4) \;=\; 2318.55 - 2251.02 \;=\; 67.53$$

Answer: The interest earned during the 5th year is \$67.53.

Question 16

During a period of 9 months, the number of people owning a flip-o-phone has increased from $285\,000$ to $315\,000$.
Calculate how many months after the start of the 9-month period we can expect $400\,000$ owners of a flip-o-phone if we assume that their number:

(a) grows according to a linear function;

(b) grows exponentially;

(c) grows according to a formula of the form:

$$F(t) = 500\,000 - a \cdot e^{bt},$$

where a and b are constants, and F denotes the number of owners of a flip-o-phone and t the time in months since the start of the 9-month period.

Tip: Linear and Exponential Growth

Let t be the time and $S(0)$ the initial value.

$$\text{linear growth:} \quad S(t) = mt + q$$

$$\text{exponential growth:} \quad S(t) = S(0) \cdot (1 + r)^t$$

Linear growth adds the same amount m each time unit, starting from q at $t = 0$. Exponential growth multiplies by the same factor $1 + r$ each time unit.

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Solution 16: (a) Growth according to a linear function

Before we start, we define the variables used in the exercise:

Variables:

$S(t)$	number of flip-o-phone owners after t months
t	time in months since the start of the period

Step 1: Find the slope.

The number of owners increased by $315\,000 - 285\,000 = 30\,000$ over 9 months:

$$m = \frac{30\,000}{9} = \frac{10\,000}{3} \quad \text{owners per month}$$

Step 2: Find the intercept and write the model.

The intercept is the initial value, $b = S(0) = 285\,000$, so the linear model is:

$$S(t) = 285\,000 + \frac{10\,000}{3}t$$

Now solve for t when $S(t) = 400\,000$:

$$400\,000 = 285\,000 + \frac{10\,000}{3}t \quad \Rightarrow \quad 115\,000 = \frac{10\,000}{3}t \quad \Rightarrow \quad t = \frac{115\,000 \cdot 3}{10\,000} = 34.5$$

Answer: After approximately 34.5 months.

Solution 16: (b) Growth according to an exponential function

Step 1: Find the growth rate r.

The model is $S(t) = 285\,000 \cdot (1+r)^t$. Use $S(9) = 315\,000$:

$$285\,000 \cdot (1+r)^9 = 315\,000 \quad \Rightarrow \quad (1+r)^9 = \frac{21}{19}$$

Take the 9-th root of both sides:

$$1+r \;=\; \left(\frac{21}{19}\right)^{1/9} \;\approx\; 1.0112 \quad\Rightarrow\quad r \;\approx\; 0.0112$$

Step 2: Solve for t when $S(t) = 400\,000$.

Isolate the exponential term:

$$400\,000 \;=\; 285\,000 \cdot (1+r)^t \quad\Rightarrow\quad (1+r)^t \;=\; \frac{80}{57}$$

Since $(1+r)^9 = \dfrac{21}{19}$, taking ln of both sides gives:

$$\frac{t}{9}\ln\left(\frac{21}{19}\right) \;=\; \ln\left(\frac{80}{57}\right) \quad\Rightarrow\quad t \;=\; \frac{9\ln(80/57)}{\ln(21/19)} \;\approx\; 30.4824$$

Answer: After approximately 30.4824 months.

Solution 16: (c) Growth according to $S(t) = 500\,000 - a \cdot e^{bt}$

Step 1: Find a using the initial condition.

Substitute $S(0) = 285\,000$ and $e^0 = 1$:

$$285\,000 \;=\; 500\,000 - a \quad\Rightarrow\quad a \;=\; 215\,000$$

Step 2: Find b using $S(9) = 315\,000$.

Substitute $t = 9$ and the value of a, then isolate the exponential term:

$$315\,000 \;=\; 500\,000 - 215\,000 \cdot e^{9b} \quad\Rightarrow\quad 215\,000 \cdot e^{9b} \;=\; 185\,000$$

$$\Rightarrow\quad e^{9b} \;=\; \frac{37}{43} \quad\Rightarrow\quad b \;=\; \frac{\ln(37/43)}{9}$$

Step 3: Solve for t when $S(t) = 400\,000$.

Substitute the known values of a and b, then isolate the exponential term:

$$400\,000 = 500\,000 - 215\,000 \cdot e^{bt} \quad \Rightarrow \quad e^{bt} = \frac{100\,000}{215\,000} = \frac{20}{43}$$

$$\Rightarrow \quad bt = \ln\left(\frac{20}{43}\right) \quad \Rightarrow \quad t = \frac{\ln(20/43)}{b} = \frac{9\ln(20/43)}{\ln(37/43)} \approx 45.8418$$

Answer: After approximately 45.8418 months.

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Question 17

In the beginning of the year 1990, there were three times as many Chinese as Europeans. The Chinese population increases every year by 2%, while the European population decreases each year by 0.5%.

When will there be four times as many Chinese as Europeans?

Tip: Exponential Growth and Decay

A quantity that changes by a constant percentage each period follows one of two formulas, where $S(0)$ is the initial value and r is the rate per period (a positive decimal).

$$\text{growth:} \quad S(t) = S(0) \cdot (1+r)^t$$

$$\text{decay:} \quad S(t) = S(0) \cdot (1-r)^t$$

The value after any time is $S(0)$ times a power of the factor per period.

Solution 17: Find when the Chinese population is four times the European population

Before we start, we define the variables used in the exercise:

Variables:

$C(t)$	Chinese population t years after 1990
$E(t)$	European population t years after 1990
t	time in years since 1990

At $t = 0$ there are three times as many Chinese as Europeans:

$$C(0) = 3E(0)$$

Step 1: Write the two population formulas.

$$C(t) = 3E(0) \cdot (1.02)^t \qquad \text{and} \qquad E(t) = E(0) \cdot (0.995)^t$$

Step 2: Set up the ratio condition and simplify.

We want the time when the Chinese population is four times the European population:

$$C(t) = 4E(t)$$

Substitute the formulas and divide both sides by $E(0)$:

$$3E(0) \cdot (1.02)^t = 4E(0) \cdot (0.995)^t \quad \Rightarrow \quad 3 \cdot (1.02)^t = 4 \cdot (0.995)^t$$

Step 3: Solve for t.

Group the two powers on one side:

$$3 \cdot (1.02)^t = 4 \cdot (0.995)^t \quad \Rightarrow \quad \left(\frac{1.02}{0.995}\right)^t = \frac{4}{3}$$

Take ln of both sides:

$$t \ln\left(\frac{1.02}{0.995}\right) = \ln\left(\frac{4}{3}\right) \quad \Rightarrow \quad t = \frac{\ln(4/3)}{\ln(1.02/0.995)} \approx 11.5930$$

Answer: Approximately 11.5930 years after 1990, i.e. around mid-2001.

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Question 18

A certain country's national income grows by 4.5% per year.

(a) Find the growth percentage per decade (i.e., period of 10 years).

(b) Find the growth factor per semester.

Tip: Growth Rate and Growth Factor

A quantity growing at a constant rate follows:

$$S(t) \;=\; S(0)\cdot(1+r)^{t}$$

Growth Percentage (per period): r

Growth Factor (per period): $1+r$

Solution 18: (a) Find the growth percentage per decade

The yearly growth factor is $1+0.045=1.045$. Apply the model over 10 years:

$$S(10) \;=\; S(0)\cdot 1.045^{10} \;\approx\; S(0)\cdot 1.5530$$

So after a decade the value is about 1.5530 times the starting value. Converting back to a percentage:

$$(1.5530-1)\cdot 100\% \;=\; 55.30\%$$

Answer: The growth percentage per decade is approximately 55.30%.

Solution 18: (b) Find the growth factor per semester

A semester is half a year. Apply the model over $\frac{1}{2}$ year:

$$S(0.5) \;=\; S(0) \cdot 1.045^{0.5} \;\approx\; S(0) \cdot 1.0223$$

Answer: The growth factor per semester is approximately 1.0223.

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Question 19

The value of a car decreases by 15% every year. How long does it take until the value of the car is half its original value?

Tip: Solving for Time in Exponential Decay

A quantity decaying at rate r follows:

$$S(t) \;=\; S(0)\cdot(1-r)^{t}$$

To solve for t, isolate the decay factor and take ln of both sides.

Solution 19: Find how long until the car is worth half its original value

Before we start, we define the variables used in the exercise:

Variables:

$S(t)$	value of the car after t years
t	time in years

The car's value decreases by 15% per year, so its decay factor is $1-0.15=0.85$:

$$S(t) \;=\; S(0)\cdot(0.85)^{t}$$

We want the time t when the value is half the original, $S(t)=0.5\cdot S(0)$:

$$S(0)\cdot(0.85)^{t} \;=\; 0.5\cdot S(0) \quad\Rightarrow\quad 0.85^{t} \;=\; 0.5$$

Take ln of both sides:

$$t\ln(0.85) \;=\; \ln(0.5) \quad\Rightarrow\quad t \;=\; \frac{\ln(0.5)}{\ln(0.85)} \;\approx\; 4.27$$

Answer: It takes approximately 4.27 years for the car's value to be halved.

Question 20

An investor has a choice of investing a sum of money using two investment plans:

- **Plan A:** The money is invested at 4% compounded annually.
- **Plan B:** The money is invested at 3.75% compounded semi-annually.

(a) Which of the two plans is more favorable to the investor?

(b) If the investor chooses Plan A, after how many years will the initial sum be increased by 50%?

(c) Suppose the investor starts with an initial sum of €30 000 and that they follow Plan A. Write down a function equation to calculate the one-year interest earned at the end of year t.

Suppose now that the investor follows a plan to invest €10 000, such that the interest function at the end of year t is given by:

$$I(t) = 10\,000 \cdot 0.076 \cdot 1.076^{t-1}.$$

Every year the investor has to pay taxes on the interest earned:

- If the interest is lower than €1,660, no taxes have to be paid.
- If the interest exceeds €1,660, a 15% tax is due on the amount exceeding €1,660.

(d) For how many years does the investor pay no taxes?

(e) After how many years will the tax be greater than €100?

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Tip: Compounding Period Conventions

In this course a compounding phrase is read literally, not the way a bank would read it. "Compounded semi-annually at rate r" means r is the full growth rate for each half-year period (each 0.5 years). Only the exponent doubles:

$$S(t) \;=\; S(0)\cdot(1+r)^{2t}$$

The rate is never halved, so the real-world form below is not used here:

$$S(t) \;=\; S(0)\cdot\left(1+\frac{r}{2}\right)^{2t}$$

In the same way, "compounded monthly" multiplies the exponent by 12:

$$S(t) \;=\; S(0)\cdot(1+r)^{12t}$$

Tip: Interest in a Specific Year

The interest earned during year t is the difference between the account value at the end of year t and the value at the end of the previous year:

$$I(t) \;=\; S(t) - S(t-1)$$

Solution 20: (a) Which of the two plans is more favorable to the investor?

Before we start, we define the variables used in the exercise:

Variables:

$A(t)$	value of the Plan A investment after t years
$B(t)$	value of the Plan B investment after t years
t	time in years

Compare the yearly growth factor of each plan.

Plan A: Annual compounding at 4% gives a yearly growth factor of:

$$1 + 0.04 \;=\; 1.04$$

Plan B: Semi-annual compounding at 3.75% means a growth factor of $1 + 0.0375$ every half-year, so the yearly growth factor is:

$$(1.0375)^2 \;\approx\; 1.0764$$

Since $1.0764 > 1.04$, Plan B grows faster.

Answer: Plan B is more favorable to the investor.

Solution 20: (b) After how many years will the initial sum under Plan A increase by 50%?

An increase of 50% means the growth factor must reach 1.5:

$$(1.04)^t \;=\; 1.5$$

Take ln of both sides:

$$t\ln(1.04) \;=\; \ln(1.5) \quad\Rightarrow\quad t \;=\; \frac{\ln(1.5)}{\ln(1.04)} \;\approx\; 10.34$$

Answer: Approximately 10.34 years.

Solution 20: (c) Write a function for the one-year interest earned under Plan A

Before we start, we define the variables used in the exercise:

Variables:

$S(t)$	value of the Plan A investment after t years
$I(t)$	interest earned during year t
t	time in years

With an initial sum of $S(0) = 30\,000$ growing at 4% per year, the account value is $S(t) = 30\,000 \cdot (1.04)^t$. Apply the interest-in-a-specific-year formula:

$$I(t) = S(t) - S(t-1) = 30\,000 \cdot (1.04)^t - 30\,000 \cdot (1.04)^{t-1}$$

$$= 30\,000 \cdot (1.04)^{t-1} \cdot (1.04 - 1) = 1\,200 \cdot (1.04)^{t-1}$$

Answer: $I(t) = 1\,200 \cdot (1.04)^{t-1}$.

Solution 20: (d) For how many years does the investor pay no taxes?

The new interest function is $I(t) = 10\,000 \cdot 0.076 \cdot 1.076^{t-1} = 760 \cdot 1.076^{t-1}$. No tax is due while $I(t) < 1\,660$:

$$760 \cdot 1.076^{t-1} < 1\,660 \quad \Rightarrow \quad 1.076^{t-1} < \frac{83}{38}$$

Take ln of both sides (the base $1.076 > 1$, so the inequality direction is unchanged):

$$(t-1)\ln(1.076) < \ln\left(\frac{83}{38}\right) \quad \Rightarrow \quad t - 1 < 10.66 \quad \Rightarrow \quad t < 11.66$$

Since t counts full years, the interest stays below the threshold for years $t = 1, 2, \ldots, 11$.

Answer: For 11 years.

Solution 20: (e) After how many years will the tax be greater than €100?

The tax is 15% of the interest exceeding €1 660, so we need:

$$0.15 \cdot (I(t) - 1\,660) > 100 \quad \Rightarrow \quad I(t) > 1\,660 + \frac{100}{0.15} \approx 2\,326.67$$

Substitute $I(t) = 760 \cdot 1.076^{t-1}$ and isolate the exponential term:

$$760 \cdot 1.076^{t-1} > 2\,326.67 \quad \Rightarrow \quad 1.076^{t-1} > 3.0614$$

Take ln of both sides:

$$(t-1)\ln(1.076) > \ln(3.0614) \quad \Rightarrow \quad t - 1 > 15.28 \quad \Rightarrow \quad t > 16.28$$

Since t counts full years, the first year the condition holds is $t = 17$.

Answer: After 17 years.

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Question 21

When a person takes a drug, one can measure the number of milligrams of the drug in a person's system after t hours. This amount is given by:

$$D(t) = A \cdot e^{Bt}.$$

(a) Assume that a person takes 20 milligrams at the beginning. Determine A.

(b) Assume that the number of milligrams decreases by 20% every hour. Determine B.

(c) Now assume that instead, the number of milligrams decreases by 20% every two hours. Determine B.

For another drug, the number of milligrams in a person's system after t hours is given by the equation:

$$D(t) = 100 \cdot 2^{-0.5t}.$$

(d) When will the number of milligrams be half of the initial amount?

(e) After how many hours will the drug have disappeared completely, according to this model?

Tip: The Continuous Exponential Model

A quantity that grows or decays continuously is modeled by:

$$S(t) = S(0) \cdot e^{kt}$$

Here $S(0)$ is the initial value and k is the continuous rate: $k > 0$ gives growth, $k < 0$ gives decay. A decaying model approaches 0 as t grows but never reaches it.

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Tip: Matching Bases

If both sides of an equation are powers of the same base, the exponents must be equal:

$$b^x = b^y \quad \Leftrightarrow \quad x = y$$

Solution 21: (a) Determine A

Before we start, we define the variables used in the exercise:

Variables:

$D(t)$	milligrams of drug in the system after t hours
t	time in hours

The initial amount at $t = 0$ is 20 milligrams, so $D(0) = A \cdot e^0 = A$:

$$A \;=\; 20$$

Answer: $A = 20$.

Solution 21: (b) Determine B for a 20% decrease every hour

A 20% decrease means 80% remains after one hour, so $D(1)/A = 0.8$:

$$e^B \;=\; 0.8 \quad \Rightarrow \quad B \;=\; \ln(0.8) \;\approx\; -0.2231$$

Answer: $B = \ln(0.8) \approx -0.2231$.

Solution 21: (c) Determine B for a 20% decrease every two hours

The same decay factor 0.8 now applies over 2 hours, so $D(2)/A = 0.8$:

$$e^{2B} = 0.8 \quad \Rightarrow \quad 2B = \ln(0.8) \quad \Rightarrow \quad B = \frac{\ln(0.8)}{2} \approx -0.1116$$

Answer: $B = \dfrac{\ln(0.8)}{2} \approx -0.1116.$

Solution 21: (d) When will the number of milligrams be half of the initial amount?

The new model is $D(t) = 100 \cdot 2^{-0.5t}$, with initial value $D(0) = 100$. We want $D(t) = 50$:

$$100 \cdot 2^{-0.5t} = 50 \quad \Rightarrow \quad 2^{-0.5t} = 0.5$$

Rewrite 0.5 as a power of 2 and match the bases:

$$2^{-0.5t} = 2^{-1} \quad \Rightarrow \quad -0.5t = -1 \quad \Rightarrow \quad t = 2$$

Answer: After 2 hours.

Solution 21: (e) After how many hours will the drug have disappeared completely?

Step 1: Write the model.

The drug amount follows:

$$D(t) = 100 \cdot 2^{-0.5t}$$

Step 2: Check whether $D(t)$ can equal 0.

The amount is fully gone only if $D(t) = 0$, which would require:

$$2^{-0.5t} = 0$$

A power of 2 is always strictly positive, so this equation has no solution.

Step 3: Describe the long-run behavior.

As t grows, the exponent $-0.5t$ becomes large and negative, so:

$$2^{-0.5t} \to 0 \quad \text{as} \quad t \to \infty$$

The amount gets arbitrarily close to 0 but never reaches it in finite time.

Answer: Never - the amount approaches 0 as $t \to \infty$ but never reaches it exactly.

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Question 22

The country Busistan has two competing newspapers: the Busistan Times and the Busistan Mail. At this moment:

- The circulation of the Busistan Mail is twice as large as the circulation of the Busistan Times.
- The circulation of the Busistan Times increases at a rate of 2% per month.
- The circulation of the Busistan Mail decreases at a rate of 12% per year.

(a) How much does the circulation of the Busistan Times increase every year? Give your answer as a percentage.

(b) When will both newspapers have the same circulation?

Tip: Growth Rate and Growth Factor

A quantity growing at a constant rate follows:

$$S(t) = S(0) \cdot (1+r)^t$$

Growth Percentage (per period): r

Growth Factor (per period): $1+r$

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Solution 22: (a) How much does the circulation of the Busistan Times increase every year?

The monthly growth factor is $1 + 0.02 = 1.02$. Apply the model over 12 months:

$$S(12) \;=\; S(0) \cdot 1.02^{12} \;\approx\; S(0) \cdot 1.268242$$

So after one year the circulation is about 1.268242 times its current value. Converting back to a percentage:

$$(1.268242 - 1) \cdot 100\% \;\approx\; 26.8242\%$$

Answer: Approximately 26.8242% per year.

Solution 22: (b) When will both newspapers have the same circulation?

Before we start, we define the variables used in the exercise:

Variables:

$T(t)$	circulation of the Busistan Times t years from now
$M(t)$	circulation of the Busistan Mail t years from now
t	time in years from now

Step 1: Write the two circulation functions.

Let $T(0)$ be the current circulation of the Times, so the Mail's current circulation is $2T(0)$.

$$T(t) \;=\; T(0) \cdot (1.268242)^t \qquad \text{and} \qquad M(t) \;=\; 2T(0) \cdot (0.88)^t$$

Step 2: Set the two functions equal and group the powers.

Divide both sides by $T(0)$:

$$(1.268242)^t \;=\; 2 \cdot (0.88)^t \quad \Rightarrow \quad \left(\frac{1.268242}{0.88}\right)^t \;=\; 2$$

Step 3: Solve for t.

Take ln of both sides:

$$t \ln\left(\frac{1.268242}{0.88}\right) = \ln(2) \quad \Rightarrow \quad t = \frac{\ln(2)}{\ln(1.268242/0.88)} \approx 1.8966$$

In months, $1.8966 \cdot 12 \approx 22.7595$.

Answer: After approximately 1.8966 years, or 22.7595 months.

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Question 23

A person has an amount of €10 000 which he wants to use as a provision for old age. He decides to invest:

- 20% of his money in an equity fund (or stock fund),
- 80% of his money in a bond fund.

It is more risky to invest in an equity fund than in a bond fund, but in the long run, equity funds have a greater return. Assume:

- The equity fund grows by 8% per year.
- The bond fund grows by 4% per year.

Let:

- $S(t)$: The money in the equity fund after t years.
- $B(t)$: The money in the bond fund after t years.

(a) How long does it take the money in the equity fund to double in value?

(b) At the time the person starts his investment, the money in the equity fund is 20% of the total capital. What percentage of the total capital is in the equity fund after 5 years?

(c) Write an equation that gives the percentage F for the total capital that is in the equity fund after t years.

(d) After how many years is 40% of the total capital in the equity fund? Write an equation and solve this equation manually, i.e., without using the special features of your graphing calculator.

(e) How long does it take for the total amount in the equity fund and the bond fund together to double in value? Write an equation and use your graphing calculator to solve it.

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Solution 23: (a) How long does it take the money in the equity fund to double in value?

Before we start, we define the variables used in the exercise:

Variables:

$S(t)$	money in the equity fund after t years
$B(t)$	money in the bond fund after t years
t	time in years

The equity fund starts with 20% of the €10 000 and grows by 8% per year:

$$S(0) = 0.20 \cdot 10\,000 = 2\,000$$

$$S(t) = 2\,000 \cdot (1.08)^t$$

The bond fund starts with the other 80% and grows by 4% per year:

$$B(0) = 0.80 \cdot 10\,000 = 8\,000$$

$$B(t) = 8\,000 \cdot (1.04)^t$$

Doubling means the equity fund reaches twice its starting value, $S(t) = 2 \cdot S(0)$:

$$2\,000 \cdot (1.08)^t = 2 \cdot 2\,000 \quad \Rightarrow \quad (1.08)^t = 2$$

Take ln of both sides:

$$t \ln(1.08) = \ln(2) \quad \Rightarrow \quad t = \frac{\ln(2)}{\ln(1.08)} \approx 9.0065$$

Answer: Approximately 9.0065 years.

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Solution 23: (b) What percentage of the total capital is in the equity fund after 5 years?

Step 1: Find the value of each fund at $t = 5$.

Substitute $t = 5$ into each fund function:

$$S(5) = 2\,000 \cdot (1.08)^5 \approx 2\,938.66$$

$$B(5) = 8\,000 \cdot (1.04)^5 \approx 9\,733.22$$

Step 2: Find the equity fund's share of the total.

Divide the equity fund value by the combined value of both funds:

$$\frac{S(5)}{S(5) + B(5)} = \frac{2\,938.66}{12\,671.88} \approx 0.2319$$

Written as a percentage, this is about 23.19%.

Answer: Approximately 23.19%.

Solution 23: (c) Write an equation for the percentage F of total capital in the equity fund

The percentage is the equity fund's value divided by the total capital:

$$F(t) = \frac{S(t)}{S(t) + B(t)} \cdot 100\%$$

Substitute the two fund functions:

$$F(t) = \frac{2\,000 \cdot (1.08)^t}{2\,000 \cdot (1.08)^t + 8\,000 \cdot (1.04)^t} \cdot 100\%$$

Answer: $F(t) = \dfrac{2\,000 \cdot (1.08)^t}{2\,000 \cdot (1.08)^t + 8\,000 \cdot (1.04)^t} \cdot 100\%.$

Solution 23: (d) After how many years is 40% of the total capital in the equity fund?

Step 1: Set the equity share equal to 0.40 and clear the denominator.

The equity fund is 40% of the total when:

$$\frac{2\,000 \cdot (1.08)^t}{2\,000 \cdot (1.08)^t + 8\,000 \cdot (1.04)^t} = 0.40$$

Multiply both sides by the denominator:

$$2\,000 \cdot (1.08)^t = 0.40 \cdot \left(2\,000 \cdot (1.08)^t + 8\,000 \cdot (1.04)^t\right)$$

$$2\,000 \cdot (1.08)^t = 800 \cdot (1.08)^t + 3\,200 \cdot (1.04)^t$$

Step 2: Group the like-base terms.

Move the $(1.08)^t$ terms to one side:

$$1\,200 \cdot (1.08)^t = 3\,200 \cdot (1.04)^t$$

Divide both sides by $1\,200 \cdot (1.04)^t$:

$$\left(\frac{1.08}{1.04}\right)^t = \frac{8}{3}$$

Step 3: Solve for t.

Take $\ln$ of both sides:

$$t \ln\left(\frac{1.08}{1.04}\right) = \ln\left(\frac{8}{3}\right) \quad \Rightarrow \quad t = \frac{\ln(8/3)}{\ln(1.08/1.04)} \approx 25.9889$$

Answer: After approximately 25.9889 years.

Solution 23: (e) How long until the total in both funds doubles in value?

Step 1: Write the total-value function.

The total in both funds after t years is the sum of the two fund functions:

$$T(t) \;=\; 2\,000 \cdot (1.08)^t + 8\,000 \cdot (1.04)^t$$

Step 2: Set up the doubling equation.

The initial total is:

$$T(0) \;=\; 2\,000 + 8\,000 \;=\; 10\,000$$

Doubling means $T(t) = 2 \cdot T(0)$:

$$2\,000 \cdot (1.08)^t + 8\,000 \cdot (1.04)^t \;=\; 20\,000$$

Step 3: Explain why this cannot be solved by hand.

The equation has two exponential terms with different bases (1.08 and 1.04). They cannot be combined into a single power, so taking ln will not isolate t. Use the numeric solver instead.

Step 4: Solve with the numeric solver.

Write the equation with the calculator variable `X` in place of t:

$$2000 \cdot 1.08^X + 8000 \cdot 1.04^X \;=\; 20000$$

Open the numeric solver (`MATH` $\rightarrow$ `Solver...`) and enter the two sides on separate rows:

$$\texttt{E1:}\quad 2000 \cdot 1.08^X + 8000 \cdot 1.04^X$$

$$\texttt{E2:}\quad 20000$$

Enter an initial guess `X` $= 14$, then press `ALPHA + ENTER` (SOLVE):

$$t \;\approx\; 14.2714$$

Answer: Approximately 14.2714 years.

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Question 24

Answer the following questions.

(a) When Lisa was born, Lisa's grandpa invested a capital of €1 000 at a compound interest of 5% yearly. At Lisa's 18th birthday, she receives the value of the investment. What is the value of this capital at her 18th birthday?

(b) In the same way, grandpa invested a capital of €1 000 at the same compound interest of 5% yearly, when Lisa's brother Greg was born. At the time of Lisa's 18th birthday, Greg's capital has grown to the value of €2 105. How old is Greg at that time?

(c) Lisa and Greg have another brother Jonathan, who is seven years younger than Lisa. Also when Jonathan was born, grandpa invested a capital of €1 000 at the compound interest of 5% yearly. Is there a time when the value of Lisa's capital is exactly one and a half times the value of Jonathan's capital? If so, calculate how old Lisa is at that time. If not, explain why not.

(d) Ahmed's grandpa invested €900 when Ahmed was born, at a fixed yearly compound interest rate. By the time Ahmed turns 18, this capital has grown to €3 021. What was the yearly fixed interest rate?

Tip: Compound Interest Formulas

Let $S(0)$ be the principal, r the annual interest rate as a decimal, and t the time in years.

$$\text{future value:} \quad S(t) = S(0) \cdot (1+r)^t$$

$$\text{total interest earned:} \quad S(t) - S(0)$$

$$\text{interest in year } t\text{:} \quad S(t) - S(t-1)$$

Solution 24: (a) Find the value of Lisa's capital at her 18th birthday

Before we start, we define the variables used in the exercise:

Variables:

$L(t)$	value of Lisa's capital after t years
t	time in years

Substitute $L(0) = 1\,000$, $r = 0.05$, and $t = 18$:

$$L(18) \;=\; 1\,000 \cdot (1.05)^{18} \;\approx\; 2\,406.62$$

Answer: Approximately €2 406.62.

Solution 24: (b) Find Greg's age when his capital reaches €2 105

Greg's capital follows $G(t) = 1\,000 \cdot (1.05)^t$. Solve $G(t) = 2\,105$ for t:

$$1\,000 \cdot (1.05)^t \;=\; 2\,105 \quad\Rightarrow\quad (1.05)^t \;=\; 2.105$$

Take ln of both sides:

$$t\ln(1.05) \;=\; \ln(2.105) \quad\Rightarrow\quad t \;=\; \frac{\ln(2.105)}{\ln(1.05)} \;\approx\; 15.2554$$

Answer: Greg is approximately 15.2554 years old.

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Solution 24: (c) Is there a time when Lisa's capital is exactly 1.5 times Jonathan's?

Before we start, we define the variables used in the exercise:

Variables:

$L(t)$	Lisa's capital when she is t years old
$J(t)$	Jonathan's capital when Lisa is t years old
t	Lisa's age in years

Since Jonathan is 7 years younger, his investment has been growing for $t-7$ years:

$$L(t) \;=\; 1\,000\cdot(1.05)^{t} \qquad \text{and} \qquad J(t) \;=\; 1\,000\cdot(1.05)^{t-7}$$

Their ratio is constant, independent of t:

$$\frac{L(t)}{J(t)} \;=\; (1.05)^{t-(t-7)} \;=\; (1.05)^{7} \;\approx\; 1.4071$$

Since this ratio is always approximately 1.41, it can never equal 1.5.

Answer: No - the ratio of their capitals is always approximately 1.41, and never 1.5.

Solution 24: (d) Find the yearly fixed interest rate

Ahmed's capital follows $A(t) = A(0)\cdot(1+r)^{t}$. Substitute $A(0)=900$, $A(18)=3\,021$, and solve for r:

$$900\cdot(1+r)^{18} \;=\; 3\,021 \quad\Rightarrow\quad (1+r)^{18} \;=\; \frac{3\,021}{900} \;\approx\; 3.3567$$

Take the 18th root of both sides:

$$1+r \;=\; (3.3567)^{1/18} \;\approx\; 1.069589 \quad\Rightarrow\quad r \;\approx\; 0.069589$$

Answer: The yearly interest rate was approximately 6.9589%.

Question 25

The GDP per capita of the United States is equal to \$28 020 and that of Chile is equal to \$4 860.

(a) Draw a graph with a logarithmic scale that shows the GDP per capita of the United States and Chile.

(b) On the graph from part (a), draw horizontal lines corresponding to a GDP per capita of \$5, \$50, \$500, \$5 000 and \$50 000. What do you notice? Explain!

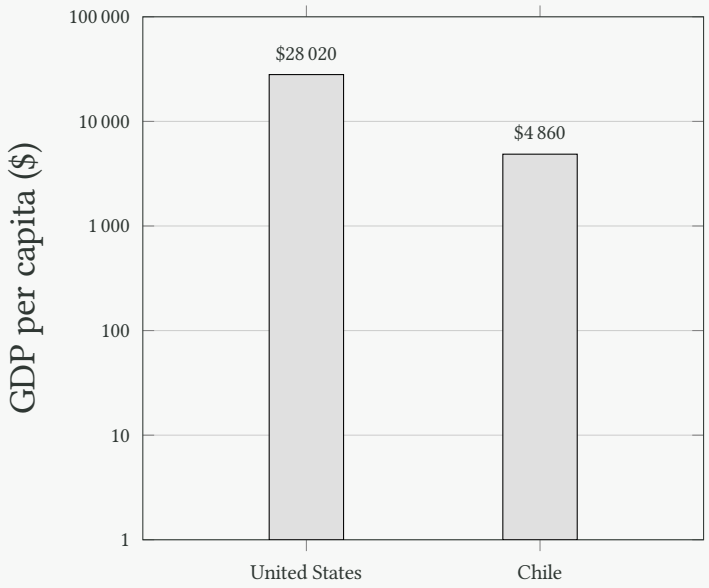
Tip: Reading a Logarithmic Scale

On a normal (linear) scale, two numbers sit an equal distance apart when they differ by the same *amount*: 10, 20, 30, ... are evenly spaced.
On a logarithmic scale, two numbers sit an equal distance apart when they differ by the same *factor*: 1, 10, 100, 1 000, ... are evenly spaced, because each one is 10 times the previous one.
This is useful when the values cover a very wide range: small values and very large values stay readable on the same axis, and steady percentage growth shows up as a straight line.

Solution 25: (a) Graph the two GDP-per-capita values on a logarithmic scale

Plot each country's GDP per capita as a bar on a vertical axis whose gridlines step by a factor of 10 (1, 10, 100, 1 000, 10 000, 100 000). On this scale the United States bar (\$28 020) ends between the 10 000 and 100 000 lines, and the Chile bar (\$4 860) ends between the 1 000 and 10 000 lines.

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Answer: Each GDP per capita is drawn as a bar on a base-10 logarithmic vertical axis, with the United States bar between the $10\,000$ and $100\,000$ gridlines and the Chile bar between the $1\,000$ and $10\,000$ gridlines.

Tip: Why Equal Factors Look Equally Spaced

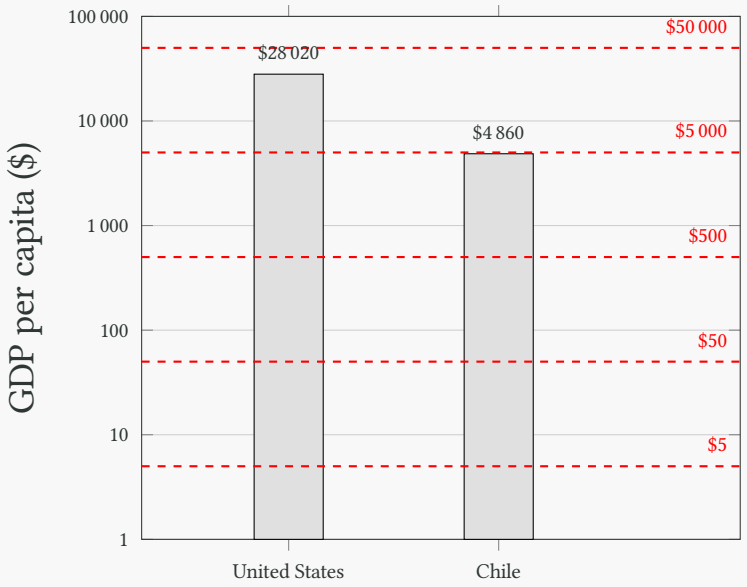
On a logarithmic axis the position of a value is set by its logarithm, and a logarithm turns multiplication into addition:

$$\log(a \cdot 10) \;=\; \log(a) + \log(10)$$

So multiplying a value by 10 always moves it up the axis by the same fixed distance, $\log(10)$.

Solution 25: (b) Add the lines at $5, $50, $500, $5 000, $50 000 - what do you notice?

Add the five values as dashed red horizontal lines on the same graph. Each one is 10 times the value below it, and on a logarithmic scale a step by a constant factor is always the same distance - so the five lines come out equally spaced.



Answer: The five dashed lines are equally spaced, because each value is 10 times the previous one and a logarithmic scale draws every $\times 10$ step as the same vertical distance.

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Question 26

A lake with a surface area of $60\,000\,\text{m}^2$ has been infected with an invasive algae species. At this moment, the algae take up only $8\,\text{m}^2$ of the lake's surface, but the area covered by the algae increases constantly. More specifically, the area covered by algae grows by a quarter every week. Let A be the surface of the pond (in m^2) that is covered by the algae after t weeks.

(a) Find an expression for the function $A(t)$.

(b) How long does it take for the algae to cover the full surface of the lake?

(c) Make a graph of the function $A(t)$.

(d) Now make a graph of the function $A(t)$ with a logarithmic scale on the vertical axis and a normal scale on the horizontal axis. What do you notice? Explain!

Tip: Exponential Growth and Decay

A quantity that changes by a constant percentage each period follows one of two formulas, where $S(0)$ is the initial value and r is the rate per period (a positive decimal).

$$\text{growth:} \quad S(t) = S(0) \cdot (1+r)^t$$

$$\text{decay:} \quad S(t) = S(0) \cdot (1-r)^t$$

The value after any time is $S(0)$ times a power of the factor per period.

Solution 26: (a) Find an expression for the function $A(t)$

Before we start, we define the variables used in the exercise:

Variables:

$A(t)$	surface area covered by algae after t weeks, in m^2
t	time in weeks

The initial area is $A(0) = 8\,\text{m}^2$. Growing "by a quarter" each week means a 25% increase, so the weekly growth factor is $1 + 0.25 = 1.25$. Substituting these values:

$$A(t) \;=\; 8 \cdot (1.25)^t$$

Answer: $A(t) = 8 \cdot (1.25)^t$.

Solution 26: (b) How long until the algae cover the full lake surface?

We want the time t when $A(t)$ equals the lake's full surface, $60\,000\,\text{m}^2$:

$$8 \cdot (1.25)^t \;=\; 60\,000 \quad\Rightarrow\quad (1.25)^t \;=\; 7\,500$$

Take ln of both sides:

$$t\ln(1.25) \;=\; \ln(7\,500) \quad\Rightarrow\quad t \;=\; \frac{\ln(7\,500)}{\ln(1.25)} \;\approx\; 39.9862$$

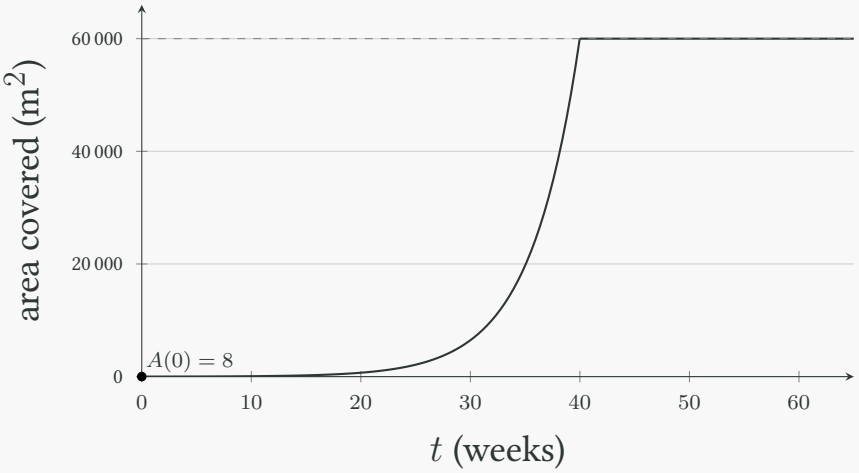
Answer: Approximately 39.9862 weeks.

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Solution 26: (c) Make a graph of the function $A(t)$

The graph starts at $A(0) = 8$ and curves steeply upward. From part (b), the algae cover the full $60\,000\,\text{m}^2$ surface at about $t \approx 40$ weeks, after which the area cannot grow any further, so the graph flattens at $60\,000\,\text{m}^2$.

Answer:



Solution 26: (d) Graph $A(t)$ with a logarithmic vertical axis - what do you notice?

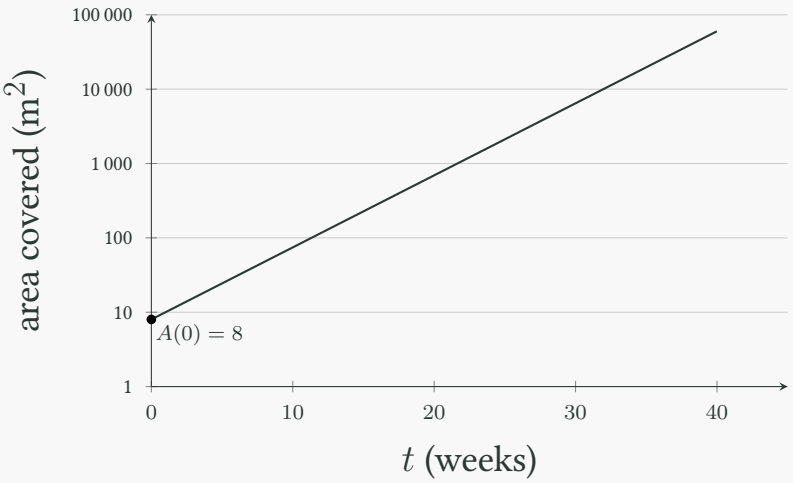
Take the natural log of both sides of $A(t) = 8 \cdot (1.25)^t$:

$$\ln(A(t)) \;=\; \ln(8) + t \cdot \ln(1.25)$$

This is a straight line in t, with intercept $\ln(8)$ and slope $\ln(1.25)$. A logarithmic vertical axis turns the constant multiplicative growth of an exponential function into constant additive steps, which is why the graph becomes a straight line.

Answer: The graph becomes a straight line, since a logarithmic scale turns constant multiplicative growth into constant additive steps.

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Question 27

14 years ago, Sean's first grandchild, Lea, was born. At the time of her birth, Sean invested a certain amount at a fixed yearly interest rate. Two years later, Lea's sibling Jordan was born. Sean invested the same amount at the same interest rate for his grandson. After this year's interest was paid, the compound amount on Lea's account is €773.47, while the compound amount on Jordan's account is €754.49.

(a) What is the fixed yearly interest rate?

(b) What is the amount originally invested?

Tip: Solving for Two Unknowns by Dividing Two Equations

When two equations share the same two unknowns and each is a constant times a power with the same base, dividing one by the other cancels the common constant and leaves a single equation in one unknown:

$$\frac{c \cdot k^{t_1}}{c \cdot k^{t_2}} = k^{t_1 - t_2}$$

Solution 27: (a) Find the fixed yearly interest rate

Before we start, we define the variables used in the exercise:

Variables:

$S(0)$	amount originally invested (the same for both accounts)
r	fixed yearly interest rate
t	time in years since the deposit

Lea is now 14 and Jordan is now 12, and each account follows $S(t) = S(0) \cdot (1 + r)^t$:

$$773.47 = S(0) \cdot (1 + r)^{14}$$

$$754.49 \;=\; S(0)\cdot(1+r)^{12}$$

Step 1: Divide the two equations to eliminate $S(0)$.

$$\frac{773.47}{754.49} \;=\; \frac{S(0)\cdot(1+r)^{14}}{S(0)\cdot(1+r)^{12}} \;=\; (1+r)^{2}$$

Step 2: Solve for r.

The left side is $773.47/754.49 \approx 1.02517$, so:

$$(1+r)^{2} \;=\; 1.02517$$

Take the square root of both sides:

$$1+r \;=\; (1.02517)^{1/2} \;\approx\; 1.0125 \quad\Rightarrow\quad r \;\approx\; 0.0125$$

Answer: The fixed yearly interest rate is 1.25%.

Solution 27: (b) Find the amount originally invested

Substitute $r = 0.0125$ into Lea's equation and solve for $S(0)$:

$$773.47 \;=\; S(0)\cdot(1.0125)^{14}$$

$$S(0) \;=\; \frac{773.47}{(1.0125)^{14}} \;=\; \frac{773.47}{1.1895} \;\approx\; 650.25$$

Answer: Approximately €650.

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Review 1 - Exponential and Logarithmic Functions

Chapter 2 - Functions

Review 2 - Functions

Chapter 3 - Derivatives

Review 3 - Derivatives

Chapter 4 - Applications of Derivatives

Review 4 - Applications of Derivatives

Exam: 2020 Example

Exam: 2020 January

Exam: 2020 September 1

Exam: 2020 September 2

Exam: 2021 January 1

Exam: 2021 January 2

Exam: 2022 Example

Exam: 2022 January

Exam: 2023 Example

Exam: 2025 January

Exam: 2026 Example

Exam: 2026 January 1

Exam: 2026 August 1

Exam: 2026 August 2

Exam: 2027 Example Midterm

Exam: 2027 Midterm 1

Exam: 2027 Midterm 2

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